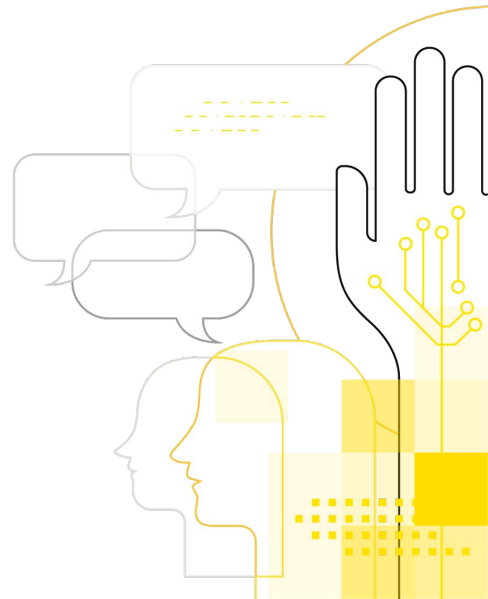
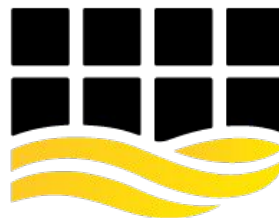


Predict Your Data in Excel with **TabH2O**



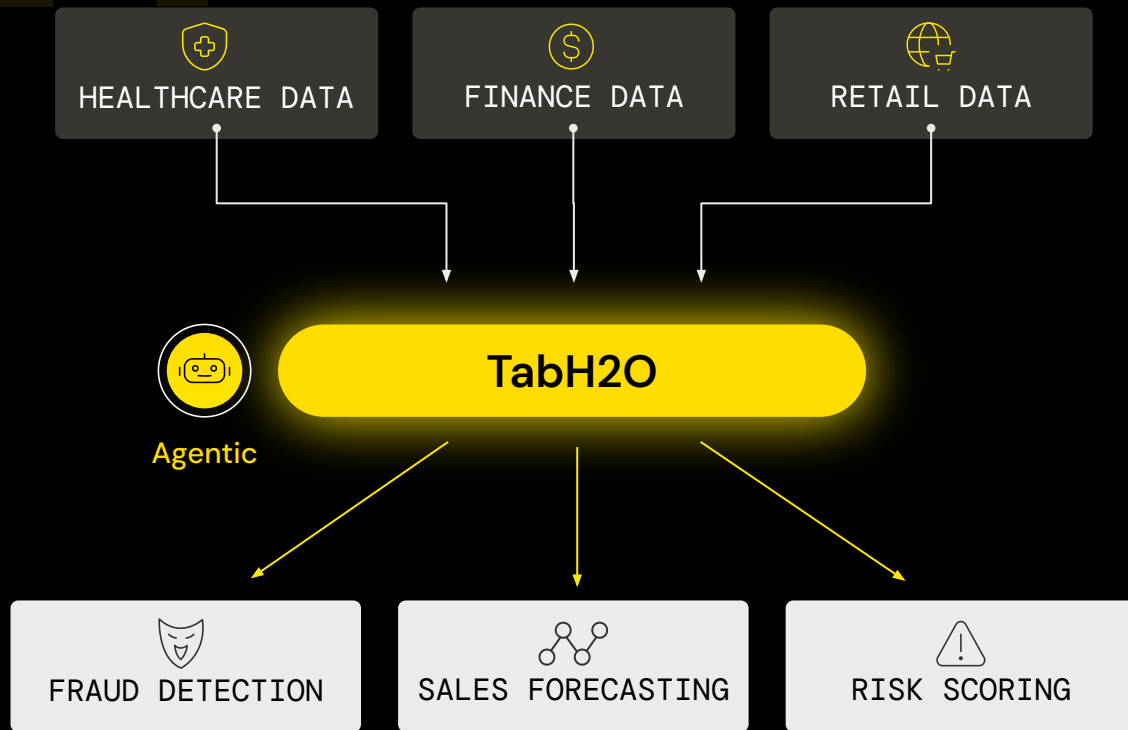
+

Rows = Historical data collected with a pre-defined unit of analysis

A 10x10 grid is shown. A horizontal brace is positioned above the first 5 columns, and a vertical brace is positioned to the left of the first 5 rows. The last column (column 10) is highlighted with a yellow border.

TabH2O > H2O.ai proprietary tabular foundation model

H2O.ai



- **Single model for all use cases**

No training, no fine-tuning, no feature engineering

- **Works across domains**

Finance, healthcare, e-commerce and many more

- **No hidden biases**

Pre-trained on millions of synthetic datasets - doesn't use semantics

- **State of the art technology**

Outperforming best in class ML models

Common Data Issues in Business Datasets



Missing & Incomplete Records



Imbalanced Data



High-Cardinality Categories



Mixed Data Types & Noisy Data



Outliers & Prediction Gaps

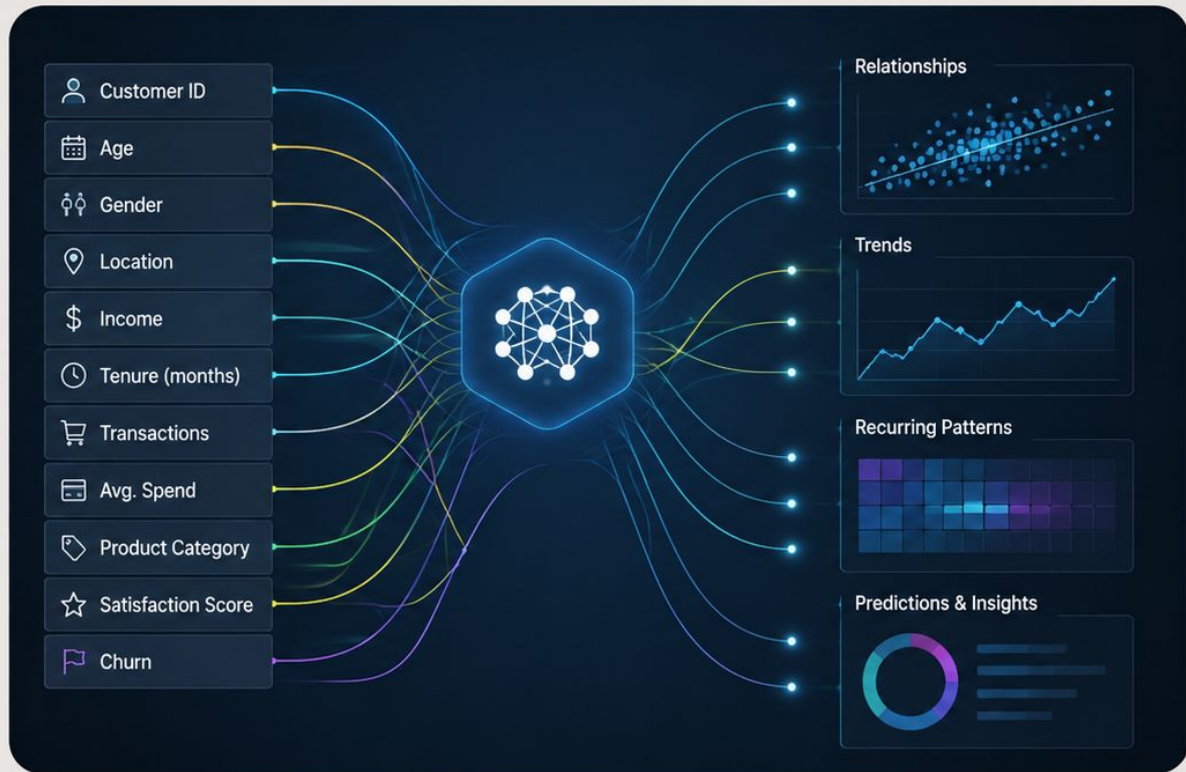
The TabH2O Advantage

Uncover patterns in imperfect business data to generate contextual predictions with **minimal preprocessing**.

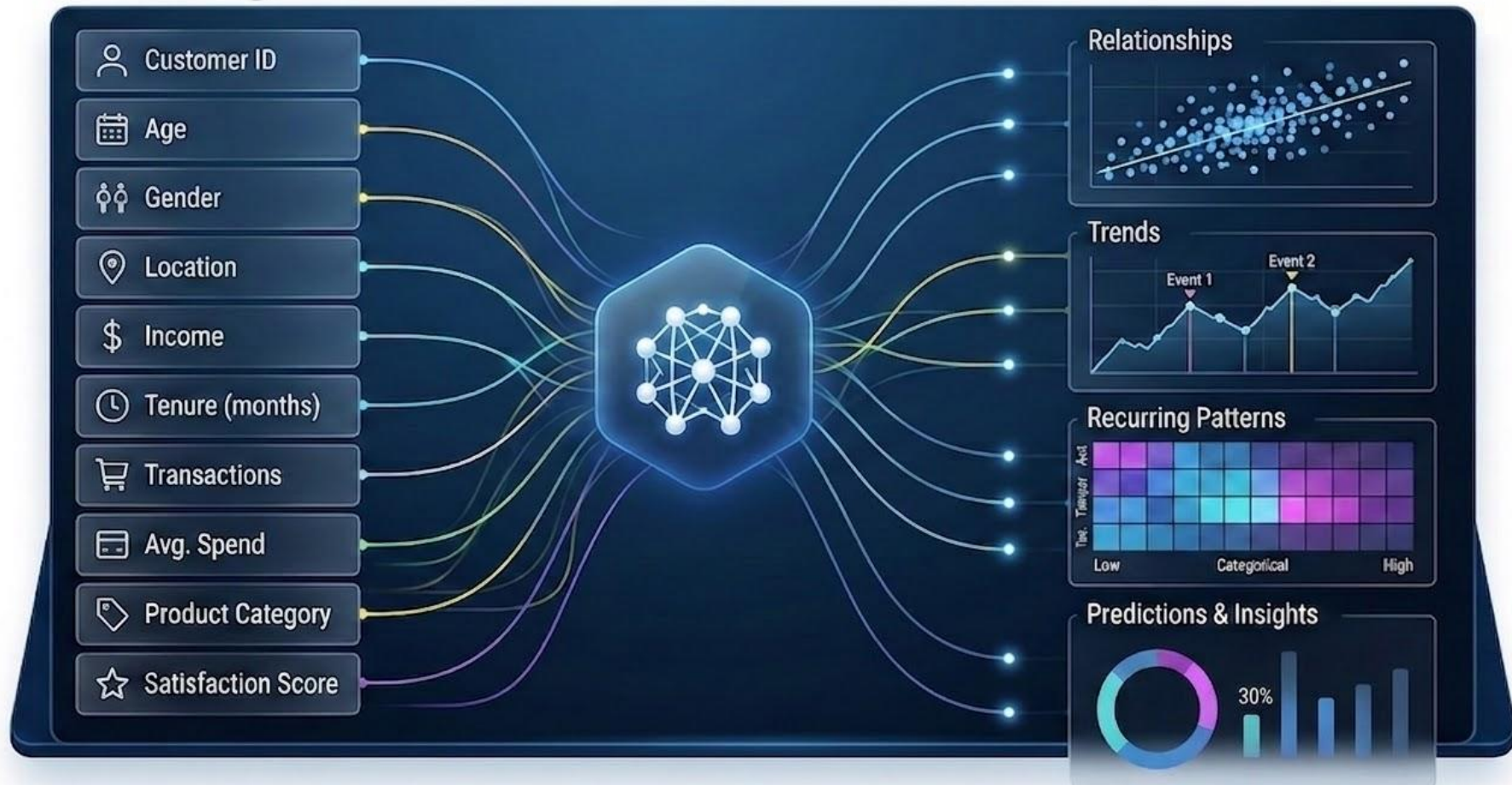
Learning Patterns from Tabular Data

TabH2O is built on a pre-trained **tabular foundation model** that has already learned common patterns found across structured datasets.

This includes **relationships** between columns, recurring **trends**, and typical behaviors often present in spreadsheets, databases, and business tables.



Learning Patterns from Tabular Data



Making Predictions: Labeled vs. Unlabeled Data



Every time you want to generate predictions, you provide a set of **labeled rows**, meaning rows where the outcome is already known, and a set of **unlabeled rows**, where the outcome is missing.

Labeled Data

(Outcome is known)

Transaction	Customer	Age	Gender	Amount	Is Fraud?
1	C001	34	F	\$120.00	No
2	C002	45	M	\$300.00	No
3	C003	29	F	\$89.00	No
4	C004	37	M	\$450.00	Yes
5	C005	52	F	\$200.00	No
...	...	...	...	...	...



Used to **train** the model
(Model learns patterns)

Unlabeled Data

(Outcome is missing)

Transaction	Customer	Age	Gender	Amount	Is Fraud?
101	C101	41	M	\$220.00	?
102	C102	28	F	\$150.00	?
103	C103	33	M	\$310.00	?
104	C104	47	F	\$95.00	?
105	C105	36	M	\$600.00	?
...	...	...	...	...	...



Machine Learning
Model



Predictions



Used to **generate predictions**
(Model applies learned patterns)



The model learns from the labeled data and applies those patterns to the unlabeled data to predict the missing outcomes.

Making Predictions:

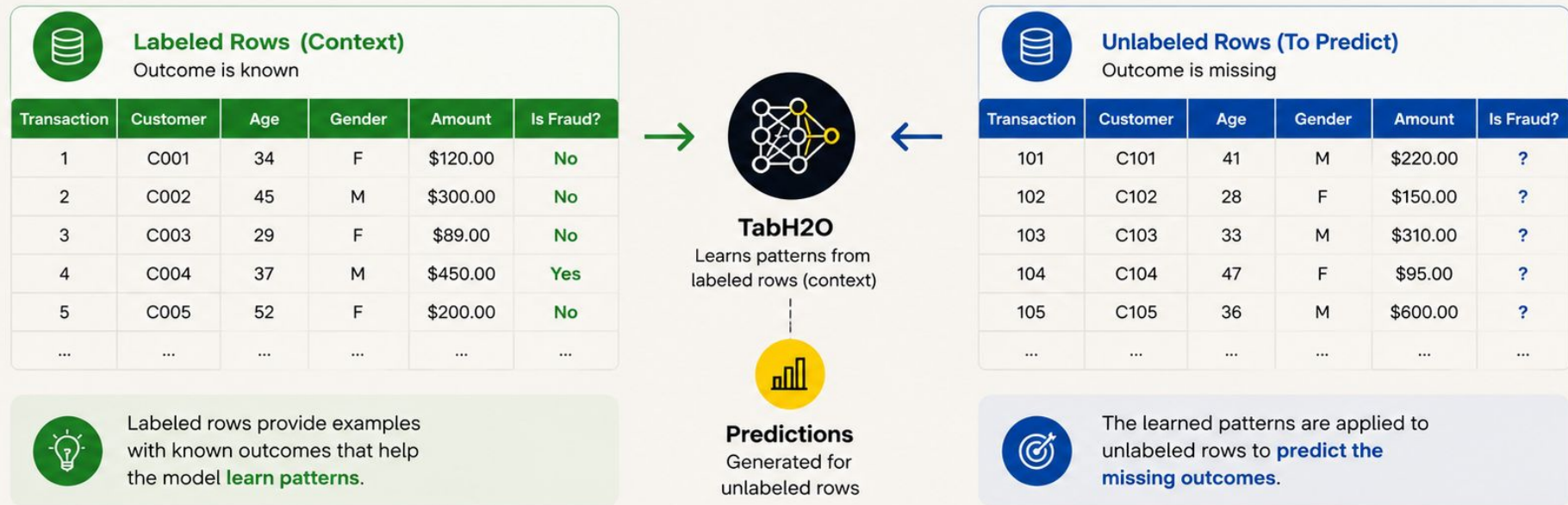
Labeled Rows vs. Unlabeled Rows

H2O.ai



TabH2O

TabH2O uses the **labeled rows as context** each time you run it to understand the patterns in your data, and **generates predictions** for the unlabeled rows accordingly.



The better and more representative your **labeled rows**, the more accurate and reliable your predictions will be.

Use Your Own Data or Follow Along with Our Sample Dataset



Use your own data

Bring your own tabular business data to test TabH2O on real-world prediction tasks.



Follow along step by step

Or use our sample customer churn dataset to reproduce the workflow demonstrated in this video.



Toy dataset available

For this demo, we've also made a small toy dataset available in our GitHub repository.

sample_customer_churn (Preview)

ID	LIMIT_BAL	SEX	EDUCATION	MARRIAGE	AGE	...	PAY_AMT1	PAY_AMT2	PAY_AMT3	PAY_AMT4	PAY_AMT5	PAY_AMT6	default payment next month (Target)
24001	50000	1	2	2	23	...	8	2401	2254	2004	704	707	0
24002	60000	1	2	2	26	...	2282	2324	2049	2000	3000	1120	1
24003	400000	1	2	2	27	...	2501	10009	1437	1105	510	959	0
24004	20000	1	5	2	27	...	0	0	0	900	0	0	0
24005	50000	1	3	2	27	...	0	100	0	70	200	100	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...
24021	360000	1	1	2	24	...	1003	1803	777	-3	-3	-3	0
24022	60000	1	1	2	25	...	5914	61210	52757	29209	26403	26969	1
24023	80000	1	2	2	25	...	0	78091	80288	82150	319	85	0



Download the latest dataset here:

github.com/h2oai/h2o-university-resources/blob/main/demo-datasets/sample_customer_churn.csv

You can find the latest download link in the description of this video.



Best Practices for Preparing Your Data in TabH2O



Use unique column names

The first row of your dataset is treated as the header, which TabH2O uses to understand the structure and meaning of your data.



Keep your dataset clean and structured

Columns that contain no values in the selected range are automatically ignored, and fully empty rows are skipped.



Headers should always be in the first row

Make sure each column contains consistent data types and meaningful values to improve prediction quality.



Selecting a range is optional

If no specific range is selected, TabH2O automatically uses the entire sheet by default.



Well-structured data leads to better results

Organized datasets help TabH2O interpret your data more reliably and generate more accurate predictions.

Key Takeaways



The first row is your header – make it count.



Empty columns are ignored, and empty rows are skipped.



Consistent, meaningful data results in better predictions.



Let TabH2O use the full sheet or select a range as needed.

Data Structure Requirements for TabH2O Predictions



If you don't select a **specific range**, TabH2O will automatically use the **entire sheet** for analysis.



Dataset Structure Rules

- ✓ The **first row** is always treated as the header.
- ✓ Each **column** should have a **clear** and **unique name**.
- ✓ Columns containing no data in the selected range are **automatically ignored**.
- ✓ Rows that are completely empty are **skipped** as well.



Prediction Requirements

For predictions to work properly, your dataset must contain **two types of rows**:



1. Labeled Rows

These rows already contain values in the target column. They help TabH2O identify patterns and relationships within your data.



2. Unlabeled Rows

These rows contain missing values in the target column. These are the rows where TabH2O will generate predictions.

ID	Age	Income	Tenure	...	default payment next month (Target)	
24001	23	50,000	2	...	0	} Labeled Rows
24002	26	60,000	3	...	1	
24003	27	40,000	1	...	0	
...	...	...	...	...	...	
24105	29	70,000	4	...	-	} Unlabeled Rows
24106	31	55,000	2	...	-	



Important Validation Rules

- ✓ You must have at least **two labeled rows**.
- ✓ Your target column must contain both **filled** and **missing** values.
- ✓ A fully **completed table** **will not work** for prediction generation.



Note:

A common mistake from first-time users is attempting to generate predictions on datasets where every target value is already filled in. In that scenario, TabH2O has no rows left to predict.

Example: Churn Use Case

In a churn prediction scenario, your dataset must contain both **labeled** and **unlabeled** rows for TabH2O to learn from existing data and generate predictions.

H2O.ai



Labeled Data (Historical Customers)

- Rows with a known churn outcome.
- Help TabH2O learn patterns and relationships in your data.



Unlabeled Data (New Customers)

- Rows without a churn outcome.
- These are the rows where TabH2O will generate predictions.

Customer_ID	Age	Tenure (Months)	Monthly Charges (\$)	Total Charges (\$)	Contract Type	Churn (Target)	
CUST_001	28	12	65.00	780.00	Month-to-month	1	Labeled (Known Outcome)
CUST_002	45	24	89.00	2,136.00	One year	0	
CUST_003	34	7	50.00	350.00	Month-to-month	1	
...	...	...	...	...	...	...	
CUST_101	30	—	60.00	—	Month-to-month	—	Unlabeled (To Be Predicted)
CUST_102	52	—	75.00	—	One year	—	
CUST_103	26	—	55.00	—	Two year	—	



Why this structure matters

Your historical customers with known outcomes act as labeled data, and new customers without outcomes are the rows you want to predict. This structure is what allows TabH2O to use your existing data as context and generate meaningful predictions for the missing values.



Summary: What You Achieved

In just a few minutes, you went from raw data to predictions—all inside your spreadsheet.



TabH2O

 What You Did	 How It Helps	 Your Benefit
 Imported your data into TabH2O	 Data is read and understood instantly	 Start faster No setup, no exports
 Let TabH2O train the model	 Patterns are learned automatically	 Save time From weeks to minutes
 Generated predictions for missing values	 Predictions are added where you need them	 Better decisions Insights at your fingertips
 Kept everything inside your spreadsheet	 No data movement, fully private	 Secure & simple No pipelines, no complexity



Your Outcome



Minutes
instead of weeks



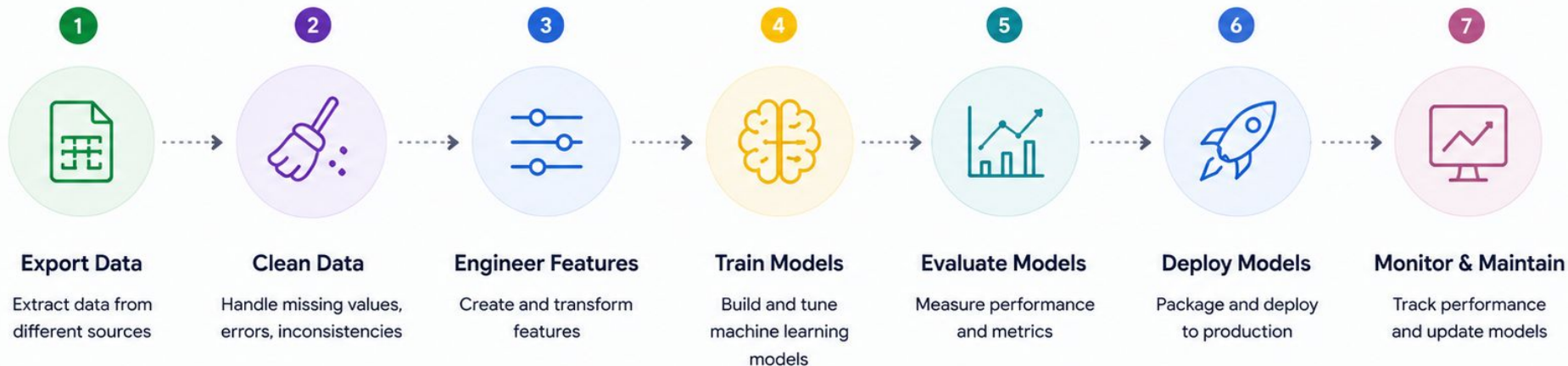
Insights
you can act on



All in one place
inside your spreadsheet

The Traditional ML Workflow

Many steps. Many tools. More complexity.



Time Consuming

From days to weeks



Requires Specialized Skills

Data, ML, and engineering expertise



Involves Multiple Tools

Disconnected workflows and environments

With **TabH2O**, everything is simpler.

H2O.ai



Stay in your spreadsheet

No exports. No pipelines.
No switching tools.



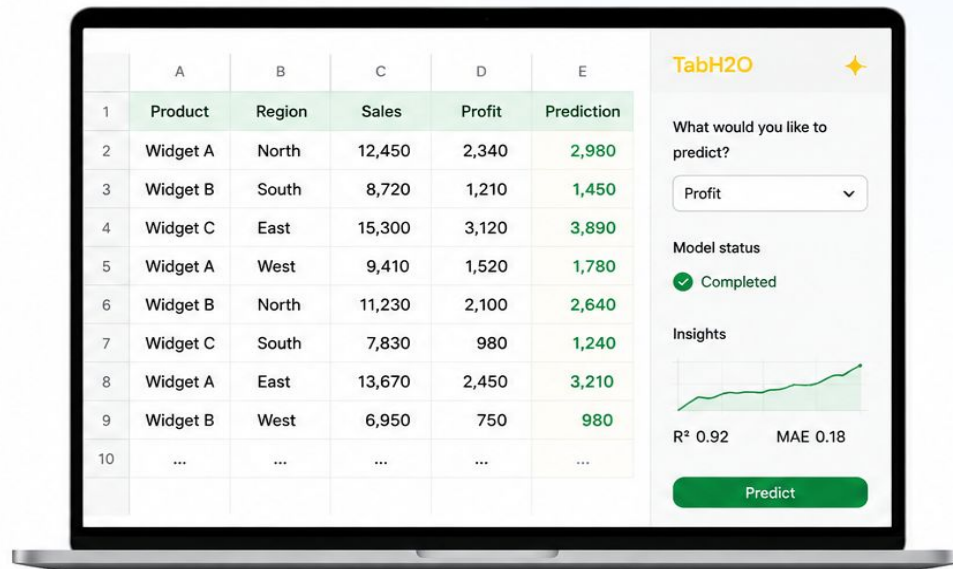
Focus on your data and your question

Ask. Predict. Understand.
All in the place you already work.



Get answers directly where you work

Predictions and insights
appear in your sheet.



Built on a foundation model approach

Delivers meaningful results even with smaller
datasets or data that is not perfectly complete.

Your data. Your control.

H2O.ai



Private by design.
Your data stays yours.



Your Data



Results

- ✓ Processed in memory
- ✓ Not stored, not cached
- ✓ Not used to retrain the model

**You stay fully in control
of your data at all times.**



Flexible and secure.
The choice is yours.

Original Data

Customer Name	City
John Smith	New York
Mary Johnson	Chicago
David Lee	Boston
...	...



Rename or
Encode (optional)

Anonymized Data

ID	Location Code
C001	NYC
C002	CHI
C003	BOS
...	...



You don't need to anonymize your data to use [TabH2O](#).



If you prefer, you can rename columns or encode values.



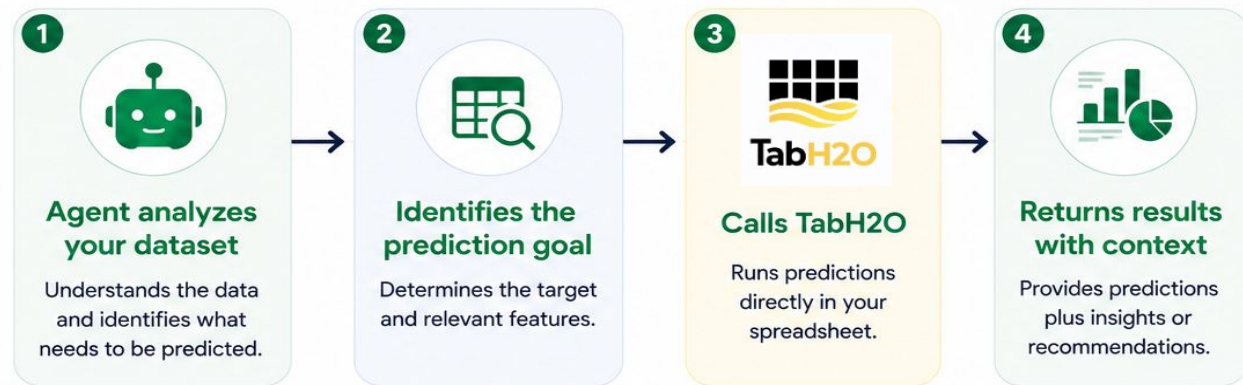
[TabH2O](#) focuses on patterns in the data, not column names – prediction quality remains the same.



Your data. Your choice. Always secure. Always in control.

Agents + TabH2O = Smarter Decisions

An agent can analyze your data, identify what needs to be predicted, call **TabH2O**, and deliver results with context or recommendations.



**More automated.
More intelligent.**

Predictions become part of a larger, data-driven decision process.



From raw data to actionable insights — all **orchestrated by your AI agent.**



Analyze



Decide



Predict



Act